# Generalized Additive Models and beyond with P-splines

## Basic information

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# Preliminary

- **Course material is available at:**

https://idaejin.github.io/courses/UPC2022/

- **Basic requirements:**
  - Basic knowledge of R software.
  - Download the latest version of R (https://www.r-project.org).
  - Rstudio (https://www.rstudio.com).
  - Regression, statistical inference.

- **Instalation of libraries**

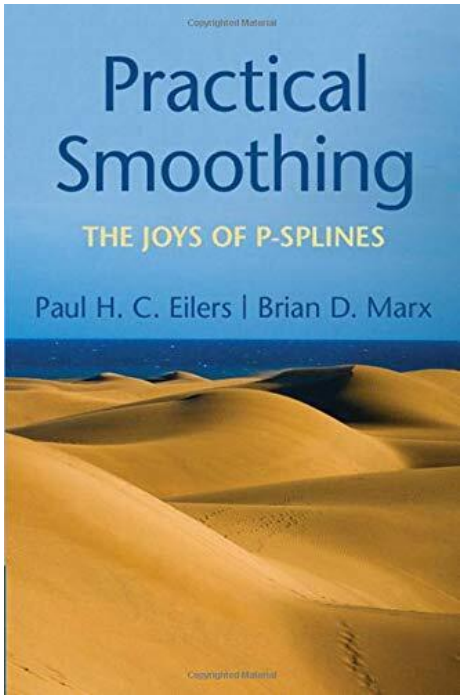
```
install.packages(c(install.packages(c("MASS","splines","nlme",
"SemiPar","mgcv","KernSmooth","gamlss","gamlss.dist","gamlss.add",
"ggplot2","lattice","latticeExtra")))
```

# Outline of this course


1. Introduction to smoothing
2. Generalized Additive Models
3. Penalized splines (P-splines)
   - 3.1 Basis and penalties
   - 3.2 Optimal smoothing
   - 3.3 P-splines for GLM's
   - 3.4 Multidimensional smoothing
4. P-splines as mixed models
   - 4.1 P-splines for GLMM's
   - 4.2 Smoothing parameters estimation as variance components
5. Beyond the exponential family
   - 5.1 Generalized Additive Mixed Models for Location, Shape and Scale
6. Examples

# Some references (I)

- "Practical Smoothing: The joy of P-splines" by Paul Eilers and Brian Marx (2021)

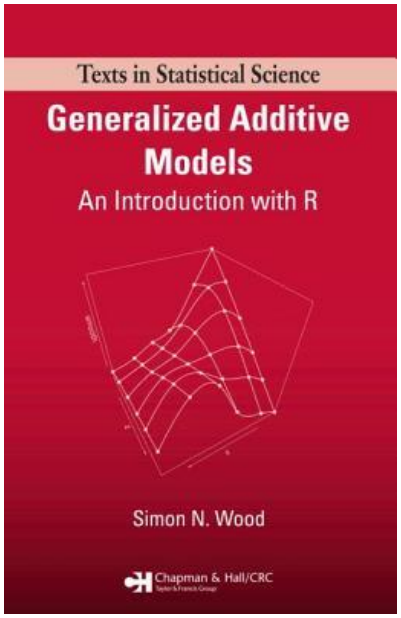



- `R` package `JOPS` in CRAN
- Highly recommended (my personal favourite)
- For beginners and practitioners
- Reproducible code to play with P-splines

- All book R code and additional material available at https://psplines.bitbucket.io/

# Some references (II)

- "Generalized Additive Models: An Introduction with R" by Simon N. Wood (2006)

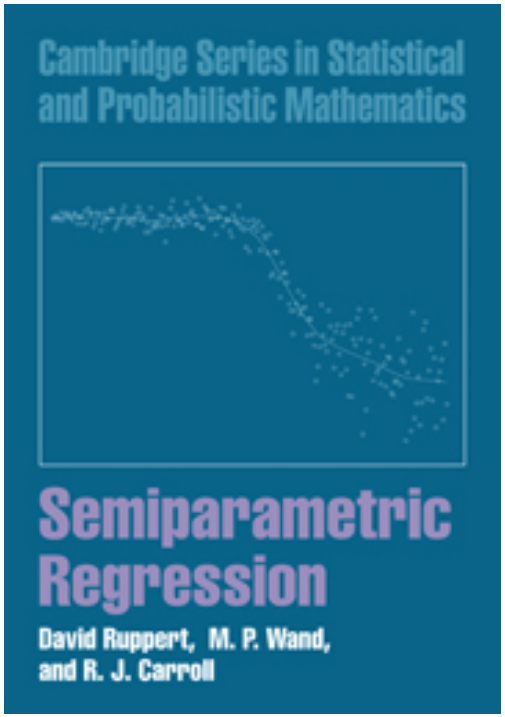



  - A very complete and well-written book
  - It covers several regression models from linear to generalized linear models, mixed models and generalized additive (mixed) models
  - For those interested in mathematical and statistical aspects (including estimation and inference)

- `R` package `mgcv` by the author, and 2nd Edition in 2017.

# Some references (III)

- "Semiparametric regression" by D. Ruppert, M.P. Wand and R.J. Carroll (2003)

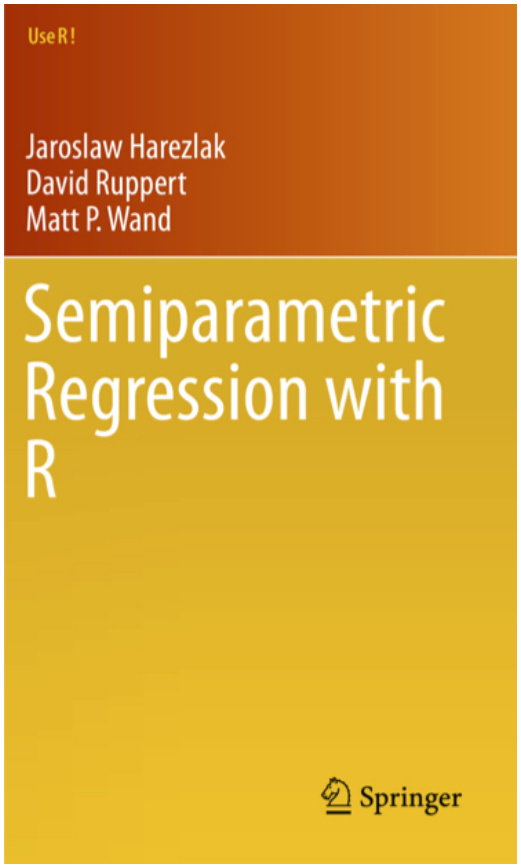



- An old classic (a little bit outdated), but still a well-written book
- Several examples and applications
- Extensive review of semi-parametric regression

- `R` package `SemiPar` by the authors (not very popular these days).

# Some references (IV)

- "Semiparametric regression with R" by J. Harezlak, D. Ruppert, M.P. Wand (2018)




- A follow-up of Semiparametric regression book of 2003
- Several examples and applications
- Extensive review of semi-parametric regression
- `R` package `HRW` by the authors.

- Website http://semiparametric-regression-with-r.net/

# Course evaluation

- **Option 1:** work with your own data (or publicly available data) and applied the GAM methodology to solve an interesting problem.
- **Option 2:** search for any publication (journal paper, manuscript, etc ...), where a GAM methodology is applied and discussed its results based on your knowledge.
- **Option 3:** Theoretical work. Find in the literature a theoretical result (not discussed in this course) and write a short report.

All these options can be discussed during the last day of the course.